Analysis Report: Business Impact of the Finova Data Breach

Finova Financial Group | January 2025 to August 2026

1. Executive Summary

In August 2026 Finova detected unauthorized access to customer contact details (email addresses and phone numbers). Core banking and transaction systems were not reported to be compromised, but the incident has raised concerns around phishing, impersonation and fraud. This report analyses 11,979 clean transactions from January 2025 to August 2026 to answer the Board's central question: who is most exposed, where is the risk concentrated, what is the financial impact and what should be prioritized.

Two definitions matter throughout this analysis. A **suspicious transaction** is one flagged by monitoring and is not confirmed fraud (SuspiciousFlag). A **fraud transaction** is a confirmed-fraud case (FraudFlag). Keeping this separate is central to every finding below.

Headline numbers:

Measure	Final value
Transactions analysed	11,979 (value ₦872.33M)
Fraud transactions	71 (fraud rate 0.59%), worth ₦4.63M (0.53% of transaction value)
Suspicious transactions	325 (2.71%), suspicious exposure ₦21.87M
Alert quality	Missed fraud 0; precision 21.85% (254 of 325 alerts were false alarms)
Exposed customers	398 of 800 (49.75%)
Revenue vs target	7.53% below target; transactions 5.75% below target

Findings:

- Fraud is small in rate but large in value. There were 71 fraud transactions in 11,979 (0.59%), worth ₦4.63M. ₦4.22M (91.04%) was in successful fraud transactions, which represents gross loss before any money was recovered.
- The fraud rate rose from 0.63% in July to 1.47% in August 2026, the highest in 20 months, in the month the breach was detected. No period after detection exists in the data, so delayed fraud cannot be measured yet.
- North Central holds 28% of fraud cases. Branches B13 (1.49%) and B17 (1.02%) are the main hotspots. Corporate customers have the highest fraud rate (1.21%), while Retail holds about 52% of fraud value, and 57% of suspicious exposure. Deposit, Loan Repayment and Bill Payment drive about 62% of fraud loss.

- Out of the 325 suspicious alerts, only 71 were confirmed fraud. No fraud was missed, but only 21.85% of the 325 suspicious alerts were confirmed fraud (254 false alarms).
- No clear link between data exposure and fraud yet**.** 398 of 800 customers (49.75%) were exposed. Their fraud rate (0.63%) is close to that of unexposed customers (0.56%). Fraud from stolen contact details often appears later, so this needs monitoring.
- Closed accounts are the highest-risk status. Their fraud rate (0.81%) is about double that of Active accounts (0.39%), and Closed with Branch has the highest suspicious rate (4.29%).
- Age: 65+ is the most affected age group (fraud rate 0.99%, with the most accurate alerts at 37.2% precision). 45-54 is second (0.66%) and has the most fraud cases (17).
- The revenue gap is not a breach effect. Revenue has been about 7.5% below target month after month, long before the breach.

Priorities for management

1. Contain the August trend with stronger live monitoring, starting in North Central.
2. Notify and protect all 398 exposed customers and monitor them for delayed fraud.
3. Close the Closed and Dormant account gap, starting with the Branch and USSD channels.
4. Audit B13, B17, B18 and B20 (rate) and B28 and B07 (value).
5. Add verification on Deposit and Loan Repayment, and extra verification for customers aged 65+ and 45-54.
6. Cut false alarms without losing the 100% catch rate, and set an early-warning trigger on monthly fraud cases.

Main limitation: the data ends in August 2026, so there is no period after the incident to measure the breach's after-effects.

2. Business Context and Objectives

2.1 Background

Finova Financial Group is a Nigerian financial services company founded in 2012 and headquartered in Lagos. It provides banking services, card services, loans and credits, and transfers to its individuals, SMEs and corporate customers across Nigeria through branches, ATMs, mobile and internet banking, USSD and POS terminals.

In August 2026 Finova detected unauthorized access to part of its customer information, including email addresses and phone numbers. The exposed data was contact information, not account credentials, so the main risk is indirect such that criminals can use phone numbers and email addresses to impersonate Finova or its customers and trick people into authorizing transactions.

Following the incident, management has observed changes in suspicious transaction activity, customer complaints, account activity, and fraud-related losses.

2.2 Objective

The main objective of this project was to investigate the business impact of the data breach and identify customers, branches, and regions most exposed to fraud, analyze suspicious transactions, assess financial losses, compare business performance with set targets and provide recommendations backed by evidence from the data.

2.3 Scope

The analysis covers all clean transactions from January 2025 to August 2026. The work followed standard analytics steps: understand the business, understand and clean the data, build a star-schema model, analyze patterns for each executive's questions, and recommend actions.

3. Questions answered:

Role	Question
CEO	What is the overall business impact of the incident?
CEO	Which customer segments are most affected?
CEO	Which regions or branches need the most attention?
CEO	Has customer activity changed following the incident?
CEO	What should management prioritize?
CRO	Which segments have the highest risk exposure?
CRO	Which channels show the most suspicious activity?
CRO	Which transaction types are most associated with fraud?
CRO	Where are the major risk hotspots?
CRO	Which risk areas need immediate monitoring?
CFO	What is the financial impact of fraudulent activity?
CFO	Which transaction types contribute most to losses?
CFO	Which regions or branches have the highest financial exposure?
CFO	How does actual performance compare with targets?
CFO	Where is the greatest opportunity to reduce losses?

4. Data Source

The data is the **DBT Case 002 dataset** from the Databeauty Tribe Business Challenge (BCS-002). It is synthetic, covers January 2025 to August 2026 and uses a single-fact-table star schema.

Table	Role	Columns	Rows
Fact_Transactions	Fact table	TransactionID, DateKey, CustomerKey, AccountKey, BranchKey, ChannelKey, TransactionTypeKey, Amount, Status, SuspiciousFlag, FraudFlag, FeeIncome	11,979
Dim_Customer	Dimension	CustomerKey, CustomerName, Segment, Status, DataExposed, Age	800
Dim_CustomerSegment	Dimension	SegmentKey, Segment (Corporate, SME, Retail, Premium)	4
Dim_Account	Dimension	AccountKey, AccountType, Status (Active, Dormant, Closed), Balance	1,000
Dim_Branch	Dimension	BranchKey, RegionKey	30
Dim_Region	Dimension	RegionKey, Region, State	10
Dim_Channel	Dimension	ChannelKey, Channel	6
Dim_TransactionType	Dimension	TransactionTypeKey, TransactionType	6
Dim_Date	Dimension	DateKey, Year, Quarter, Month (plus MonthNumber, YearMonth)	608
Targets	Monthly targets	Month, RevenueTarget, ProfitTarget, TransactionTarget, MaxFraudCases	20

Points to note:

- SuspiciousFlag does not mean confirmed fraud; FraudFlag indicates confirmed fraud.
- Revenue is measured with FeeIncome and financial exposure with Amount, as the data dictionary specifies.
- Fraud amount and all other fraud values consider every transaction status (Successful, Pending, Reversed and Failed), and no status was excluded.
- The final dataset covers 11,979 transactions, 800 customers and 20 months.

5. Tools Used

Tool	How it was used
Power BI Desktop: Power Query	Data cleaning and Validation
Power BI Desktop: DAX	Measures, and 3-page dashboard.

6. Data Preparation

Every check and fix were carried out using Power query in Power BI. The step-by-step record is in the separate Data Cleaning Documentation.

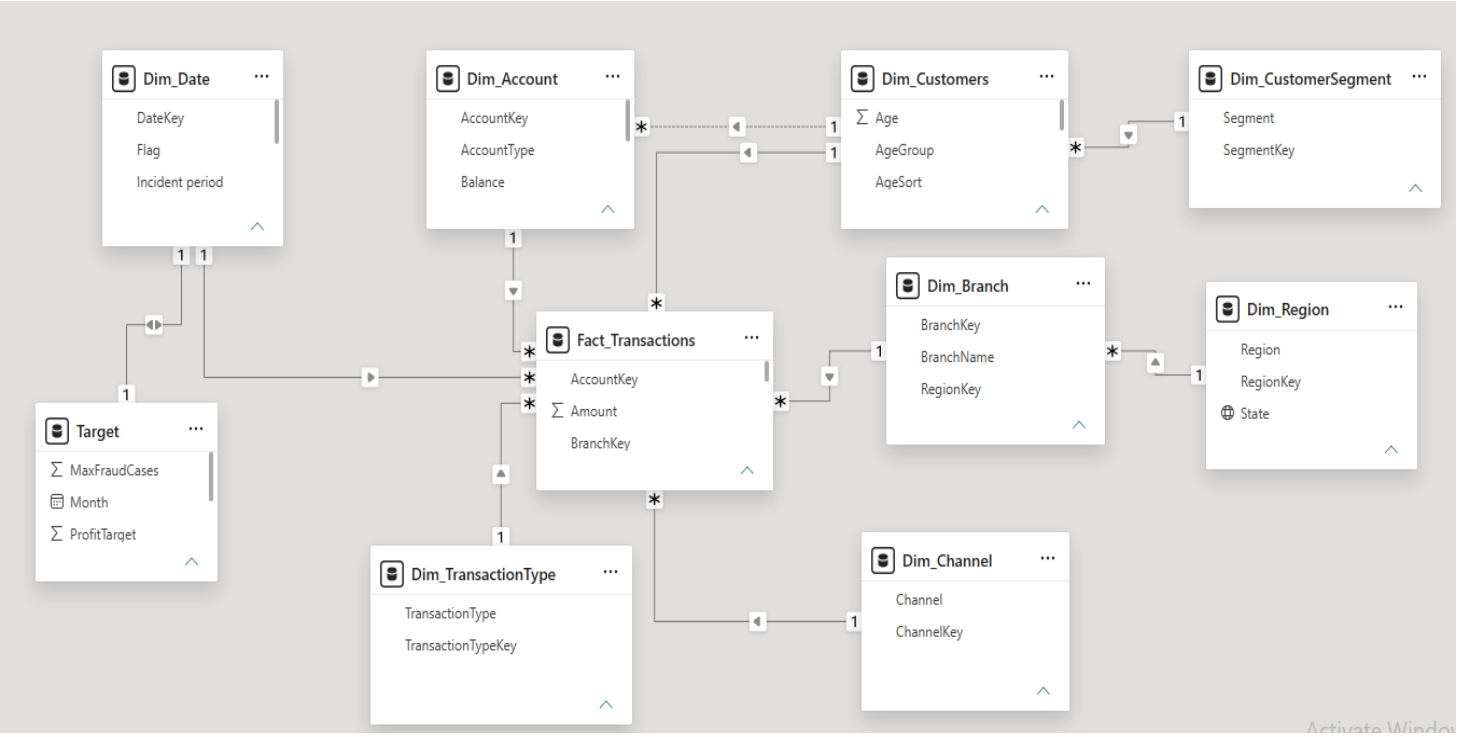
Check	Finding	Decision
Duplicates	5 identical duplicate TransactionIDs	Removed: 12,005 to 12,000 rows
Missing values	21 rows with a null Amount, all with both flags = No	**Deleted** because they are negligible (0.18%): 12,000 to **11,979** rows. Fraud (71) and suspicious (325) counts and total value were unchanged
Outliers	899 transactions (about 7.5%) by the IQR test	Kept: large transactions are normal in banking and may be the fraud signal
Orphan keys	None (Left Anti joins to every dimension)	No action
Dates	All between January 2025 and 31 August 2026	No action
Flags	Only Yes and No	No action
Targets	20 monthly rows, no issues	No action
Data types and spaces	Keys as text, numbers typed, text trimmed	Corrected
Derived fields	Age needed grouping	Created AgeGroup (six brackets), MonthNumber and YearMonth

Validated result: 11,979 transactions, 71 fraud transactions, 325 suspicious transactions, 398 exposed customers out of 800 (49.75%), and 584 customers who made at least one transaction. These figures are used consistently across the dashboard and this report. Deleting the null rows reduced total FeeIncome by ₦7,915.91 (about 0.09%) and left total transaction value unchanged.

7. Data Modelling

7.1 Star Schema

Fact_Transactions sits at the center of the schema. The dimensions relate to it one-to-many with single-direction filtering from dimension to fact, so a transaction can be sliced by date, account, branch, channel and transaction type. Four relationships extend the star so that attributes roll up logically.



7.2 Key definitions

Term	Definition
Suspicious transaction	A transaction flagged by monitoring (SuspiciousFlag). Not confirmed fraud.
Fraud transaction	A confirmed-fraud case in this challenge (FraudFlag).
Fraud rate / Suspicious rate	Fraud (or suspicious) transactions divided by total transactions.
Suspicious exposure	Total value of suspicious transactions.
Fraud amount	Total value of fraud transactions across all transaction statuses (exposure, not net loss).
Precision	Share of suspicious alerts that turned out to be confirmed fraud (fraud divided by suspicious).
Missed fraud	Fraud transactions that were not flagged as suspicious.
Exposed customer	A customer whose data was exposed in the breach (DataExposed = Yes).
AgeGroup	Six groups: 18-24, 25-34, 35-44, 45-54, 55-64 and 65+.
Gross loss	Value of fraud transactions with transaction status = Successful.

8. Analysis and findings

8.1 CEO: business impact

- Fraud is small in rate but large in value. There were 71 fraud transactions in 11,979 (0.59%), worth ₦4.63M (0.53% of ₦872.33M transaction value). The cases are concentrated, but the amount is exposure, not net loss. ₦4.22M (91.04%) was in successful transactions, which represents gross loss before any money was recovered.
- Corporate is the riskiest segment by rate; Retail is the riskiest by value.

Segment	Fraud rate	Suspicious rate	Share of fraud value	Suspicious exposure
Corporate	1.21%	3.33%	15% (₦0.69M)	₦2.2M
SME	0.60%	2.82%	24%	₦4.6M
Retail	0.57%	2.64%	52% (₦2.41M)	₦12.4M
Premium	0.31%	2.49%	9%	₦2.7M

- North Central is the main hotspot, with Branches 13 and 17 standing out. It has 20 of the 71 fraud cases (28%) and about 37% of fraud value (₦1.71M). B13 (1.49%), B17 (1.02%), B18 (0.75%) and B20 (0.74%) are the top-4 branches by rate. Branches 13 and 17 also rank in the top-4 by fraud value.

Branch	B13	B17	B18	B20	B27	B24	B22	B03	B29	B01
Fraud rate	1.49%	1.02%	0.75%	0.74%	0.53%	0.52%	0.52%	0.51%	0.26%	0.25%

- Customer activity was steady, but fraud rose in the month the breach was detected. Transaction volume held at about 600 transactions a month, but the fraud rate rose from 0.63% in July to 1.47% in August 2026 (9 cases), the highest of the 20 months. The data ends in August, the month of detection, so there is no period after the incident to measure customer behaviour afterwards.
- Data exposure shows no clear link to fraud yet. 398 of 800 customers (49.75%) had their data exposed. Their fraud rate (0.63%) is barely above that of unexposed customers (0.56%), and their suspicious rate is lower (2.63% against 2.80%). Contact details are often misused weeks or months later, so the full impact may not be visible yet.

8.2 CRO: Risk

- All 71 fraud cases were flagged, no missed fraud, but only 21.85% of the 325 alerts were confirmed fraud, leaving 254 false alarms. Precision is 21.85%, meaning that for every real fraud, about 4.6 alerts were seen. The alerts work, but they are costly for investigators.
- Retail has the highest exposure; Corporate the highest rate. Retail holds ₦12.4M of the ₦21.87M suspicious exposure (57%) because it has the highest volume (6,895 transactions). Corporate leads on rate (3.33%) but is smallest in value (₦2.2M).

- Branch is the riskiest channel. It holds the highest suspicious activity (3.27%) with about 0.73% fraud rate.

Channel	Suspicious transactions	Suspicious rate
Branch	67	3.27%
Mobile App	55	2.75%
USSD	53	2.75%
Internet Banking	52	2.55%
ATM	50	2.54%
POS	48	2.40%

- Fraud rates by transaction type sit in a narrow range. It runs from Deposit (0.70%) through Bill Payment and Cash Withdrawal (0.64% each), Loan Repayment (0.61%) and Transfer (0.52%) to Card Payment (0.45%). The gap is narrow, so no single type stands out sharply.
- Major risk hotspots include: By Region (North Central); By Branches (13 and B17); By Segment (Corporate and Retail); By channel (Branch, USSD and Mobile app), By Account status (Closed accounts).
- Closed accounts carry the highest risk. They have double the fraud rate of active accounts and they hold a large portion of the transaction volume. The hottest combinations are Account status and Channel: Closed + Branch (4.29%), Closed + USSD (3.58%) and Closed + ATM (3.26%).

Account status	Fraud cases	Fraud rate	Transactions
Closed	34	0.81%	4,178
Dormant	23	0.54%	4,251
Active	14	0.39%	3,550

- 65+ is the most affected group. It has the highest fraud rate (0.99%) and the most accurate system alerts (37.2% precision), followed by 45-54 age group with the most fraud cases (17) and fraud rate of (0.66%). Meanwhile, the 55-64 age group gets flagged the most, with suspicious rate of 3.12% but has an average fraud rate of 0.58%, meaning that the system is generating a lot of false alarms.

Age group	Customers (transacted)	Fraud cases	Fraud rate	Suspicious alerts	Suspicious rate	Precision
18-24	92	11	0.57%	54	2.81%	20.4%
25-34	94	8	0.40%	52	2.61%	15.4%
35-44	91	7	0.39%	38	2.11%	18.4%
45-54	118	17	0.66%	73	2.84%	23.3%
55-64	104	12	0.58%	65	3.12%	18.5%
65+	85	16	0.99%	43	2.67%	37.2%

8.3 CFO: Financial Impact

- Total Fraud value is ₦4.63M, which is about 0.53% of transaction value. ₦4.22M (91.04%, 63 of 71 cases) went through successful transactions, meaning that the money left the account. This is gross loss before recoveries, and only 8 cases (₦410k, 8.96%) were either stopped, reversed or still pending.
- Three transaction types drive about 62% of fraud loss: Deposit (₦971k), Loan Repayment (₦877k), and Bill Payment (₦784k). Most fraud in every transaction type went through as successful transactions. By value, 100% of Loan Repayment and Card Payment fraud was successful, followed by Bill Payment (98%) and Deposit (89%). Only a small share of the fraud in these types were stopped, reversed, or left pending.
- North Central (₦1.71M) and South South (₦1.01M) together hold about 59%. The top-4 branches (B13, B28, B07, B17) hold about 34% of fraud value.
- The largest opportunity to reduce losses is North Central Retail. As a share of total fraud value, North Central Retail holds 21.06%, followed by South South Corporate (10.20%) and South East Retail (10.17%). The top three combinations account for about 41% of fraud value.
- The revenue gap is not a breach effect. Total revenue is ₦8.36M, which is 7.53% below target (about ₦680k short) and transactions are 5.75% below, so the gap comes mostly from volume. The gap has stayed near -7.5% every month since January 2025. Fraud cases total 71% of the allowed maximum, and August's 9 cases reached 75% of its limit of 12.

Month	Revenue (₦)	Revenue vs target	Fraud cases	Cases vs max	Transactions vs target
2026-08	395,682.03	-7.53%	9	75.00%	-6.15%
2026-07	445,908.59	-7.45%	4	80.00%	-5.37%
2026-06	411,616.35	-7.52%	2	66.67%	-6.31%
2026-05	416,207.60	-7.47%	5	71.43%	-4.92%
2026-04	367,293.86	-8.04%	2	66.67%	-5.34%
2026-03	377,931.59	-8.04%	3	75.00%	-6.56%
2026-02	373,655.70	-7.41%	3	75.00%	-5.96%
Total (Jan 2025 to Aug 2026)	8,357,067.72	-7.53%	71	71.00%	-5.75%

9. Recommendations

Recommendations are listed in order of priority, based on risk and business impact, and each is tied to evidence from the analysis.

S/N	Recommendation	Evidence	Directed to
1	Contain the August trend. Strengthen live monitoring and add extra checks before approving high-risk transactions, starting with North Central.	Fraud rate rose to 1.47% in August (highest in 20 months). 91% of fraud value went through as successful transactions; North Central holds 28% of cases.	CEO

S/N	Recommendation	Evidence	Directed to
2	Notify all 398 exposed customers with a phishing and impersonation warning, offer credential resets as a precaution and monitor their accounts closely for delayed misuse.	49.75% of customers exposed; misuse of contact details often comes late, so no visible gap yet does not mean no risk.	CEO
3	Control Closed and Dormant account activity. First confirm whether these accounts were already Closed or Dormant on the transaction date, then block or step up verification, starting with Branch and USSD.	Closed fraud rate 0.81% against 0.39% for Active; Closed with Branch 4.29%, Closed with USSD 3.58%.	CRO
4	Audit the highest-risk branches: B13 and B17 first, then B18 and B20 (highest rates), and B28 and B07 (high value). Prioritise North Central.	B13 fraud rate 1.49%, about 2.5 times the average; B13, B28, B07 and B17 hold about a third of fraud value.	CRO and CFO
5	Add verification on the most affected transaction types, particularly Deposit and Loan Repayment.	Together about 43% of fraud value (Bill Payment adds another 17%).	CFO
6	Apply stronger identity verification and regular reviews for Corporate customers and closer value monitoring for Retail, starting with North Central Retail, South South Corporate and South East Retail.	Corporate fraud rate 1.21%; Retail about 52% of fraud value; the top three region-segment cells hold about 41%.	CEO and CFO
7	Add extra verification for customers aged 65+ and 45-54, such as call-back confirmation on high-value transactions.	65+ fraud rate 0.99% (about 1.7 times the average); 45-54 has the most fraud cases (17) and a 0.66% rate.	CRO
8	Reduce false alarms without losing coverage. Rank alerts by how often they turn out to be fraud (for example Closed with Branch). Keep rules tight for 65+, where alerts work, and review the rules for 55-64 and 25-34. Before changing any rule, re-run it on past transactions to confirm all 71 fraud cases would still be flagged.	Precision 21.85% and 254 false alarms; precision 37.2% for 65+, 15.4% for 25-34; 55-64 flagged the most (3.12%) with an average fraud rate.	CRO
9	Add an early-warning trigger, when monthly fraud cases reach about 80% of the monthly maximum.	Months ran at 67% to 80% of their limit, so an 80% trigger would only have fired in July.	CFO
10	Track fraud rates of exposed against unexposed customers for the next several months to measure delayed fraud .	The data ends in the month of detection.	CEO and CRO
11	Treat the revenue gap separately. Address volume and growth against target as a commercial issue, not as an effect of the data breach.	The gap has been about -7.5% month after month, before the breach; transactions are 5.75% below target.	CFO

10. Conclusion

The analysis showed that although confirmed fraud accounted for a small share of transactions (0.59%), ₦4.63M was exposed and ₦4.22M of it went through as successful transactions, representing gross loss before recoveries. The rise in fraud rate in August 2026 highlights the need for stronger monitoring. The team recommends controls before approving high-risk transactions, closer monitoring of high-risk branches, accounts and transaction types, and continued protection of the 398 exposed customers. Further analysis is needed to understand the full impact of the data breach and determine whether it contributed to changes in fraud activity or business performance.

11. Limitations and assumptions

- The data is synthetic and ends in August 2026, so there is no period after the incident to measure the breach's after-effects.
- Account status is held once per account and may differ from the account's status on a transaction's date.
- Age group customer counts reflect the 584 customers who transacted; 216 of the 800 customers had no transactions in the period.
- Profit target was not analyzed because the data contains no cost information from which profit could be calculated.